

Direct and Indirect Salt Effects on Homotypic Phase Separation

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Abstract

The low-complexity domain of hnRNPA1 (A1-LCD) phase separates in a salt-dependent manner. Unlike many intrinsically disordered proteins (IDPs) whose phase separation is suppressed by increasing salt concentrations, the phase separation of A1-LCD is promoted by > 100 mM NaCl. To investigate the atypical salt effect on A1-LCD phase separation, we carried out all-atom molecular dynamics simulations of systems comprising multiple A1-LCD chains at NaCl concentrations from 50 to 1000 mM NaCl. The ions occupy first-shell as well as more distant sites around the IDP chains, with Arg sidechains and backbone carbonyls the favored partners of Cl⁻ and Na⁺, respectively. They play two direct roles in driving A1-LCD condensation. The first is to neutralize the high net charge of the protein (+9) by an excess of bound Cl⁻ over Na⁺; the second is to bridge between A1-LCD chains, thereby fortifying the intermolecular interaction networks in the dense phase. At high concentrations, NaCl also indirectly strengthens π - π , cation- π , and amino- π interactions, by drawing water away from the interaction partners. Therefore, at low salt, A1-LCD is prevented from phase separation by net charge repulsion; at intermediate concentrations, NaCl neutralizes enough of the net charge while also bridging IDP chains to drive phase separation. This drive becomes even stronger at high salt due to strengthened π -type interactions. Based on this understanding, four classes of salt dependence of IDP phase separation can be predicted from amino-acid composition.

eLife assessment

In this potentially **important** study, the authors conducted atomistic simulations to probe the salt-dependent phase separation of the low-complexity domain of hnRNPA1 (A1-LCD). The authors have identified both direct and indirect mechanisms of salt modulation, provided explanations for four distinct classes of salt dependence, and proposed a model for predicting protein properties from amino acid composition. There is a range of opinions regarding the strength of evidence, with some considering the evidence as **incomplete** due to the limitations in the length and complexity of the atomistic MD simulations. The work should be put into a better context in relation to previous studies of salt effects on protein phase separation.

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Introduction

Biomolecular condensates formed via liquid-liquid phase separation (LLPS) mediate a variety of cellular functions such as biogenesis of the ribosome and stress response.^{1,2} The driver for phase separation or condensation is intermolecular interactions, including electrostatic, hydrogen bonding, π - π , cation- π , amino- π , and hydrophobic.^{3–5} Salt can tune all these interactions and thus exert significant effects on phase separation. While the screening effect of salt on electrostatic interactions is well-known, its effects on other types of interactions may be indirect and perhaps are less appreciated. In particular, high salt strengthens hydrophobic interactions by increasing the surface tension of water.^{6,7} Because salt can exert disparate effects on different types of interactions, it can be used as a perturbation to dissect the relative importance of these interactions in phase separation.^{8,9} Inside cells, proteins can encounter varying salt conditions at different locations or at different times, and therefore the drive for their phase separation can span a wide range.

Numerous studies of salt effects on protein phase separation have been reported. The most typical effect is the suppression of phase separation by screening electrostatic attraction, both for homotypic systems^{10–17} and for heterotypic systems.^{8,18} However, salt can also promote phase separation, although the mechanism is not always clear.^{10,19–27} In particular, Krainer et. al.¹⁹ observed a “reentrant” salt effect on the phase separation of five intrinsically disordered proteins (IDPs): phase separation occurs without salt, is prohibited by medium salt, and reemerges at high salt. They attributed the reemergence of phase separation to strengthened π -type and hydrophobic interactions that overcompensate weakened electrostatic attraction. A unifying understanding of how salt affects the phase separation of IDPs is still lacking. For example, it is an open question whether salt effects on phase separation can be predicted from the protein sequence. Deep knowledge, in particular at the atomic level, of how salt affects intermolecular interactions and, ultimately, phase separation is required.

The low-complexity domain of hnRNPA1 (A1-LCD) represents another IDP where an atypical salt effect was reported.¹⁰ The full-length protein, comprising folded domains along with the LCD, phase separates in the absence of salt and the tendency to phase separate is reduced upon adding salt, thus exhibiting the typical salt effect. A screening mechanism, specifically of electrostatic attraction between the LCD and the folded domains, is supported by small-angle X-ray scattering and coarse-grained molecular dynamics (MD) simulations. In contrast, A1-LCD does not phase separate without salt and starts to do so only after 100 mM NaCl is added. An earlier study revealed that π -types of interactions, mediated by aromatic residues, drive the phase separation of A1-LCD.²⁸ A follow-up study, based on charge mutations, further showed that the net charge plays a strong suppressive role.²⁹ The saturation concentration (C_{sat}) for phase separation is minimum at a net charge near 0, and increases by two orders of magnitude, signifying an enormous weakening of the drive for phase separation, when the net charge moves away from neutrality in either direction. In a more recent study, salt promoted the homotypic phase separation of both A1-LCD and FUS-LCD, but suppressed the heterotypic phase separation of their mixture.⁸ The salt effect on the phase separation of A1-LCD was modeled by a Debye-Hückel potential in coarse-grained simulations.³⁰ Similarly, the salt effect on the phase separation of another IDP, Ddx4, was analyzed using the random phase approximation based on a coarse-grained representation.³¹ Coarse-grained simulations with explicit water and ions have been used to study salt effects in both homotypic and heterotypic phase separation.³²

All-atom MD simulations can uniquely provide mechanistic insight into the driving force and properties of biomolecular condensates.⁵ For example, these simulations showed that ATP, a small molecule with a -4 charge, bridges between positively charged IDP chains in driving phase separation.³³ The intermolecular interactions quickly break and reform, explaining why the

condensates can rapidly fuse despite very high macroscopic viscosity. Similarly, quick breakup and reformation of salt bridges in a heterotypic condensate allow the protein molecules to be extremely dynamic in a highly viscous environment.¹⁸ Recently, all-atom MD simulations provided explanations for wide variations in phase equilibrium and material properties among condensates of tetrapeptides with different amino-acid compositions.³⁴ These and other simulations³⁵ show that all attractive residue-residue contacts contribute to the drive for phase separation.

Here we study salt effects on A1-LCD condensation by all-atom MD simulations. The simulations reveal two direct effects and one indirect effect of NaCl: neutralization of net charge and bridging between protein chains as well as strengthening of π -type interactions by drawing water away from the interaction partners. We also present a unified picture of salt dependences of phase separation by defining four distinct classes and predict these classes from amino-acid composition.

Results

Salt condenses A1-LCD and increases inter-chain interactions

The 131-residue A1-LCD is comprised mostly of Gly and Ser (51 and 22, respectively), followed by 18 aromatic residues (11 Phe and 7 Tyr), 17 residues with sidechain amides (13 Asn and 4 Gln), and 15 charged residues (10 Arg, 2 Lys, and 3 Asp), with a large net charge of +9 (**Figure 1A**). Using initial conformations of A1-LCD from the previous single-copy simulations,³⁶ we built an 8-copy model for the dense phase (with a concentration of 3.5 mM) at five NaCl concentrations ranging from 50 to 1000 mM (**Figure 1B**). Each system was simulated in four replicates for 1 μ s each. All the results reported here are averages over the four replicate simulations. From the same initial configuration with very few inter-chain contacts, the systems expand slightly at 50 mM NaCl (**Figure 1C**) but condense noticeably at 1000 mM NaCl (**Figure 1D**). At low salt, the protein chains fill the simulation box in two of the three orthogonal directions, with chain configurations rearranging frequently (Supplementary Figure 1). In contrast, at high salt, the chains aggregate into a sphere with much slower chain reconfiguration (correlation time increasing from 30 to 150 ns).

We also quantified the contrast between low and high salt by calculating D_{max} , the maximum side length of the rectangular box that parallels the simulation box and circumscribes the multi-chain system. The D_{max} values are above or close to the initial value (~ 150 Å) at 50 and 300 mM NaCl, but show a clear decrease (to ~ 125 Å) at 500 and 1000 mM NaCl (**Figure 2A**). The D_{max} results at 150 mM NaCl are mixed, falling between those at 50 and 300 mM NaCl as expected for the first 400 ns and the last 100 ns but dipping below those at 300 mM for the other half of the simulation time. The latter out-of-order behavior reflects the fact that the multi-chain system can sometimes get stuck in an overly condensed state. Overall, the systems show growing condensation with increasing salt. A small part of the reason for the growing condensation is the compaction of the individual chains. The average radius of gyration (R_g) shows a decreasing trend with increasing salt (**Figure 2B**), which matches the experimental data.¹⁰ Correspondingly, the average number of intrachain contacts per residue increases slightly, by 4%, when the salt concentration is increased from 50 mM to 1000 mM (**Figure 2C**). However, the main driver of the condensation is inter-chain interactions, with inter-chain contacts per residue increasing by 22% over the same salt range. Clearly, high salt induces A1-LCD condensation, with an increased number of interactions between protein chains. Below we present the molecular mechanisms for these effects.

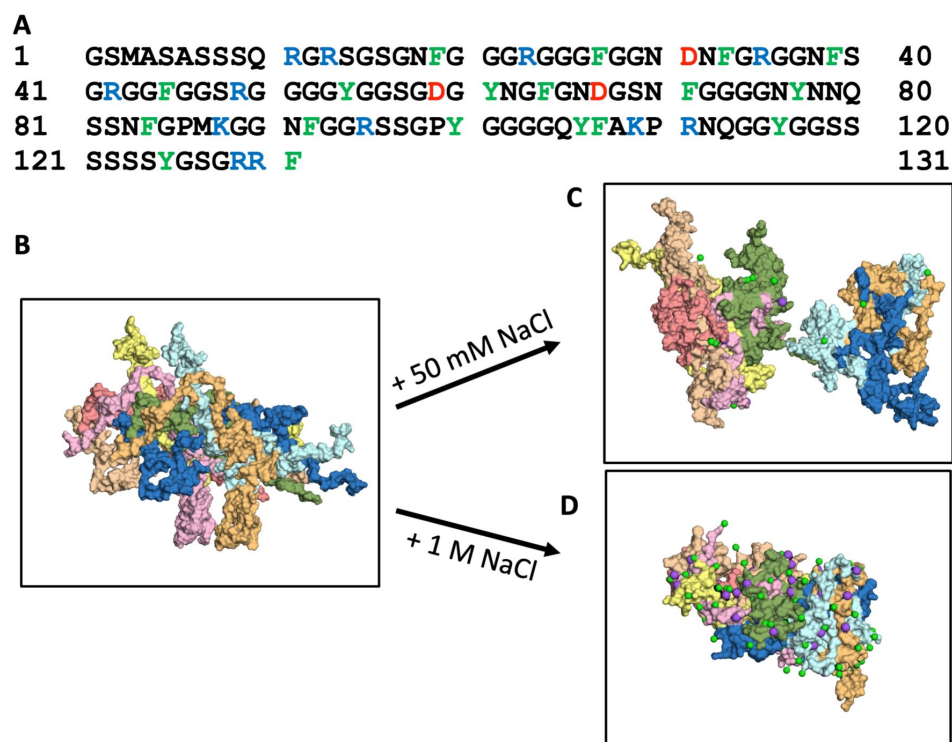


Figure 1.

Amino-acid sequence of A1-LCD and molecular dynamics simulations of its condensation. (A) Amino-acid sequence. (B) First and (C, D) last frames from 1- μ s simulations of the 8-chain systems at low and high salt. In all figures, Cl^- or Na^+ ions are represented by green and magenta spheres, respectively.

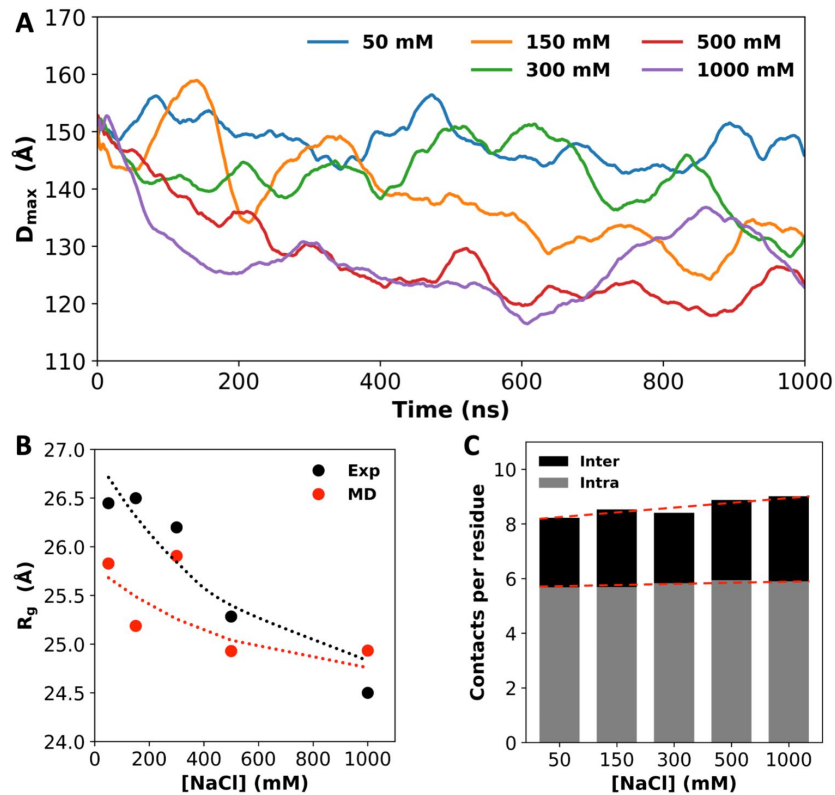


Figure 2.

Salt effects on A1-LCD condensation and inter-chain interactions. (A) $D_{\max}$ values, averaged over four replicates, as a function of simulation time. (B) Radii of gyration (R_g) from small-angle X-ray scattering¹⁰ and from MD simulations. Dashed curves are drawn to guide the eye. (C) Average number of inter- or intra-chain contacts per residue at each salt concentration. Dashed lines are drawn to show trends.

Ions selectively bind to backbone and sidechain sites to neutralize the net charge of A1-LCD

The large net charge of A1-LCD implies significant electrostatic repulsion between chains. Potentially salt ions can neutralize the net charge. To investigate ion-protein binding, we calculated radial distribution functions (RDFs) of Cl^- and Na^+ around polar groups of A1-LCD. At 1000 mM NaCl, the RDFs of Cl^- show a strong 1st peak at 3.2 Å around Arg and Lys sidechain nitrogens and a moderate 1st peak around the Gln and Asn sidechain nitrogens and the Ser sidechain oxygen (Supplementary Figures 2A). Each Arg sidechain often coordinates two Cl^- ions simultaneously, but each Lys sidechain coordinates only one Cl^- ion. A 2nd peak, at ~5 Å, is also observed around Arg and Lys sidechain nitrogens. Of the remaining polar groups, only the Cl^- RDFs around the Tyr sidechain oxygen and the backbone nitrogen show a weak 1st peak (Supplementary Figures 2B). We used cutoffs of 4 and 6.4 Å, respectively, to define 1st-shell and 2nd-shell Cl^- binding. On a per-residue basis, Arg and Lys sidechains coordinate the most 1st-shell Cl^- ions, reaching ~0.25 ions per residue (**Figure 3A**, left panel). In comparison, each Gln, Asn, or Ser residue, on average, coordinates ~0.05 Cl^- ions. Given the large numbers of Arg and Ser residues in the A1-LCD sequence, these two residue types coordinate the most 1st-shell Cl^- ions, 21.6 and 8.9, respectively, in the 8-chain system at 1000 mM NaCl. Although the RDF around backbone nitrogens has only a weak 1st peak, given their large number (131 per chain), they actually coordinate 14.1, or the second most 1st-shell Cl^- ions. Among these 14.1 Cl^- ions, 57% are coordinated with Gly residues. A large part of this high percentage is due to the enrichment of Gly (39% of all residues) in the A1-LCD sequence, but Gly is additionally favored for its lack of sidechain, which allows the close approach of ions to the backbone.

Na^+ shows a very strong preference for the Asp carboxyl in 1st-shell coordination, with a peak RDF value of 12.5 at 2.2 Å (Supplementary Figure 3A). Each Asp sidechain carboxyl typically coordinates a single Na^+ ion, usually in a bifurcated geometry. In addition, Na^+ shows a strong preference in 1st-shell coordination with Asn and Gln sidechain amide oxygens and the backbone carbonyl oxygen as well as a moderate preference with the Ser sidechain oxygen. A 2nd peak, at ~4.2 Å, is also observed in the Na^+ RDF around the Asp carboxyl. Of the remaining polar groups, only a weak 1st peak is seen in the RDF around the Tyr sidechain oxygen (Supplementary Figure 3B). We used cutoffs of 3 and 5.4 Å, respectively, to define 1st-shell and 2nd-shell Na^+ binding. On a per-residue basis, the Asp sidechain coordinates the most 1st-shell Na^+ ions, reaching

0.22 ions per residue (**Figure 3A**, right panel). The per-residue Na^+ ion number reduces to ~0.06 for Asn and Glu sidechain oxygens and further to 0.03 for the backbone oxygen and Ser sidechain oxygen. Again, due to their large number, backbone oxygens coordinate a very large number, 30.5, of 1st-shell Na^+ in the 8-chain system at 1000 mM NaCl. This number dwarfs the counterparts for sidechains, the largest of which are 7.8, 5.3, and 4.6, respectively, for Asn, Asp, and Ser. For coordination with the backbone, similar to the case with Cl^- , 55% of the 30.5 Na^+ ions involve Gly residues, again reflecting the fact that this amino acid allows the close approach of ions to the backbone. On the other hand, the largest number of 1st-shell Cl^- ions are coordinated with Arg sidechains but the largest number of 1st-shell Na^+ ions are coordinated with backbone carbonyls. Whereas each Arg sidechain often coordinates two Cl^- ions, multiple backbone carbonyls often coordinate a single Na^+ ion (see below).

In **Figure 3B**, we display the total numbers of 1st-shell Cl^- and Na^+ ions in the 8-chain systems as a function of NaCl concentration. Both the 1st-shell Cl^- and Na^+ ions increase with increasing NaCl concentration, and Cl^- ions always outnumber Na^+ ions. The difference remains almost constant, with around 6 more 1st-shell Cl^- ions than 1st-shell Na^+ ions. This difference is not enough to neutralize the total charge, +72, on the protein chains. However, when 2nd-shell ions are also included, the excess of bound Cl^- ions over bound Na^+ ions grows quickly with increasing NaCl

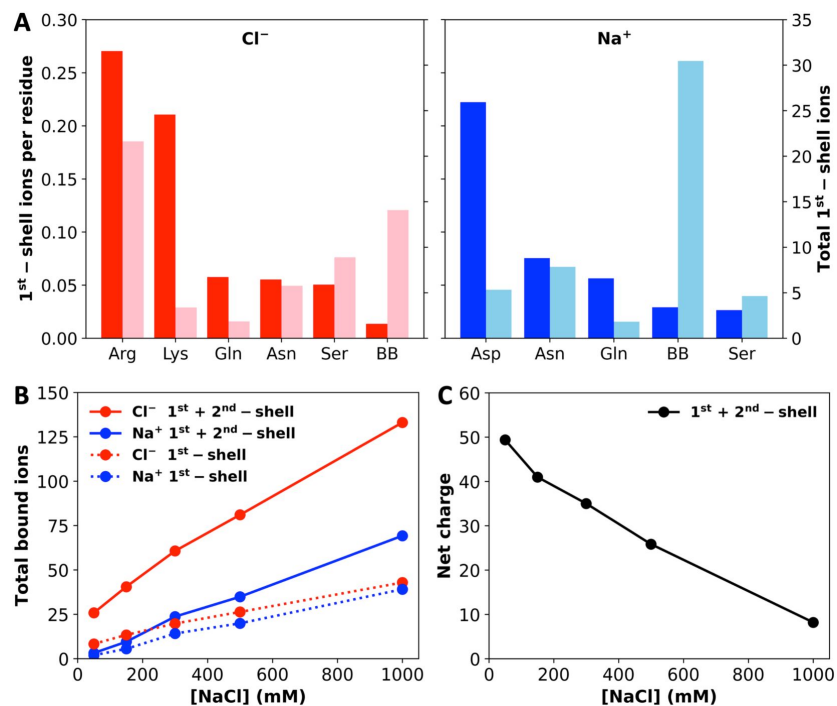


Figure 3.

Levels of ion binding at different NaCl concentrations. (A) Number of 1st-shell ions, with (darker colors; tick marks on the left vertical axis) and without (lighter colors; tick marks on the right vertical axis) normalization by the number of residues of a given amino-acid type, at 1000 mM NaCl. (B) Total number of 1st-shell only or 1st- and 2nd-shell ions. (C) Net charge of the system with 1st- and 2nd-shell ions included.

concentration. Correspondingly, the net charge of the system reduces to only +8 at 1000 mM NaCl (**Figure 3C**), which may be completely neutralized by ions beyond the second shell. Net charge repulsion explains the expansion of the chains in the simulations at 50 mM NaCl (**Figure 1C**) and the absence of phase separation at this low salt concentration.¹⁰ The protein net charge is largely neutralized by 1st- and 2nd-shell ions at high salt; hence the protein chains condense (**Figure 1D**) and phase separation is readily observed.¹⁰

Ions act as bridges between protein chains to drive condensation

In addition to charge neutralization, we suspected that ions could also fortify intermolecular interactions by bridging between proteins, similar to the role played by ATP molecules in driving phase separation of positively charged IDPs.³³ Indeed, we found that Cl⁻ has a tendency to bind with Arg and other sidechains from multiple chains and likewise, Na⁺ has a tendency to bind with backbone carbonyls and sidechain oxygens from multiple chains (**Figure 4A**).

To quantify this tendency, we calculated the number of bridging Cl⁻ or Na⁺ ions, i.e., those that bind (both 1st- and 2nd-shell) to residues on more than one A1-LCD chain (**Figure 4B**). Close to 20% of all 1st- and 2nd-shell ions bridge between A1-LCD chains. For both Cl⁻ and Na⁺, the average number of bridging ions increases with increasing salt, but the pace of increase is much greater for Na⁺ than for Cl⁻, commensurate with the trend shown by the total number of bound ions of each type (**Figure 3B**). The number of bridging ions is 4.1 for Cl⁻ and only 0.7 for Na⁺ in the 8-chain system at 50 mM NaCl, and increases to 24.4 and 12.0, respectively, at 1000 mM NaCl. The greater fold change of bridging Na⁺ ions (18-fold, vs 6-fold for Cl⁻) is also apparent when we break the bridging ions according to the number of A1-LCD chains being bridged (**Figure 4C**). For Cl⁻, the curves plotting the ion count against the number of bridged chains are close to each other among the different NaCl concentrations, but the counterparts for Na⁺ are more spread out.

These disparate effects of salt concentration on chain bridging by Cl⁻ and Na⁺ can be explained by the difference in coordination properties between the two ion types presented above. Cl⁻ strongly prefers Arg sidechains and, even at low salt, occupies a large number of sites lined by them, of which ~20% are bridging sites. However, there is a relatively limited supply of Arg sidechains (a total of 80 in the 8-chain system) and hence the fold change in bridging Cl⁻ ions upon salt increase is somewhat tempered. In contrast, although Na⁺ prefers Asp sidechains, there are very few of those in the system; instead, Na⁺ predominantly binds to backbone carbonyls. As the latter binding is relatively weak, the number of bridging Na⁺ ions is very small at low salt. However, since there is a large supply of backbone carbonyls (a total of 1048 in the 8-chain system), at high salt, Na⁺ ions bind to a portion of these backbone carbonyls and bridge A1-LCD chains. Consequently, at low salt, chain bridging is dominated by Cl⁻ (Na⁺ only 14% of bridging ions), but at high salt, Na⁺ becomes more even (33% of bridging ions) with Cl⁻ in bridging A1-LCD chains. An important reason for the latter is Na⁺ coordination by backbone carbonyls, especially those of Gly residues (**Figure 4A**).

Salt also contributes to condensation indirectly by strengthening π -type interactions

As noted above, the number of inter-chain interactions increases by 22% when the NaCl concentration increases from 50 mM to 1000 mM. While this result could be accounted for by the two direct salt effects presented so far, i.e., charge neutralization and chain bridging, which act to condense A1-LCD, there are additional factors. A breakdown of inter-chain sidechain interactions into different types reveals that when the salt concentration is increased from 50 to 1000 mM, the number of salt bridges per chain actually decreases by 7%, while the numbers of π - π , cation- π , and amino- π interactions increase by 37%, 27%, and 21%, respectively (**Figure 5A**). That is, as the A1-LCD is condensed at high salt, there is an overall increase in inter-chain interactions, but there is also a redistribution of these interactions among the different types. At increasing salt, more π -types of interactions are formed while fewer salt bridges are formed, suggesting a strengthening of the former interactions but a weakening of the latter.

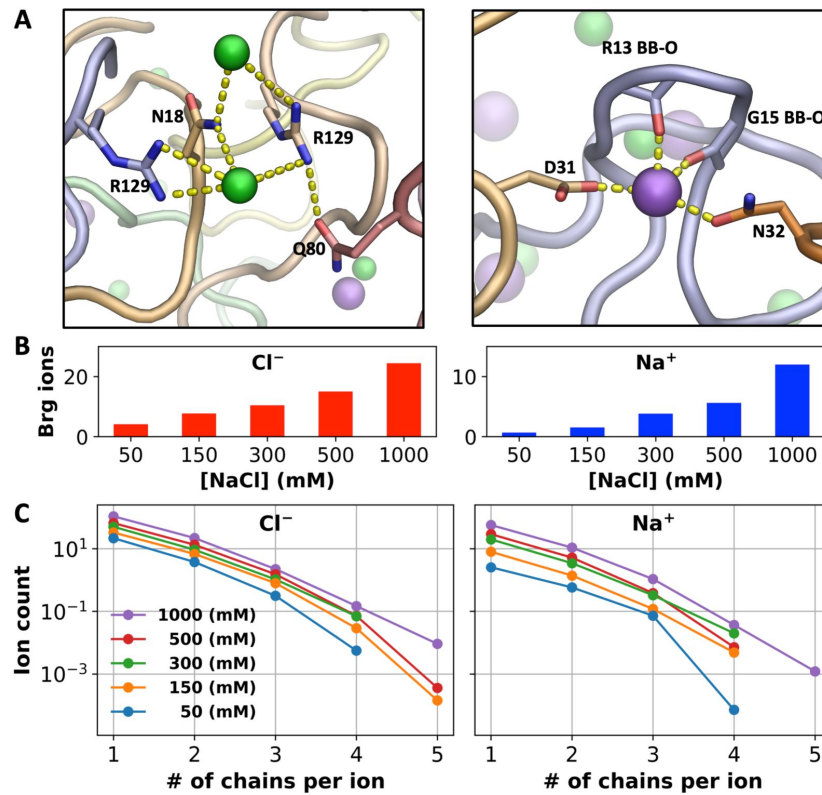


Figure 4.

Bridging ions. (A) Examples of chain bridging by Cl⁻ and Na⁺; Cl⁻ ions are coordinated to Arg and other sidechains whereas Na⁺ ions are coordinated to both backbone carbonyls (including from Gly) and sidechain oxygens. (B) Average number of Cl⁻ or Na⁺ ions engaged in bridging between A1-LCD chains. (C) Average number of ions bound in 1st- and 2nd-shell sites lined by a given number of A1-LCD chains.

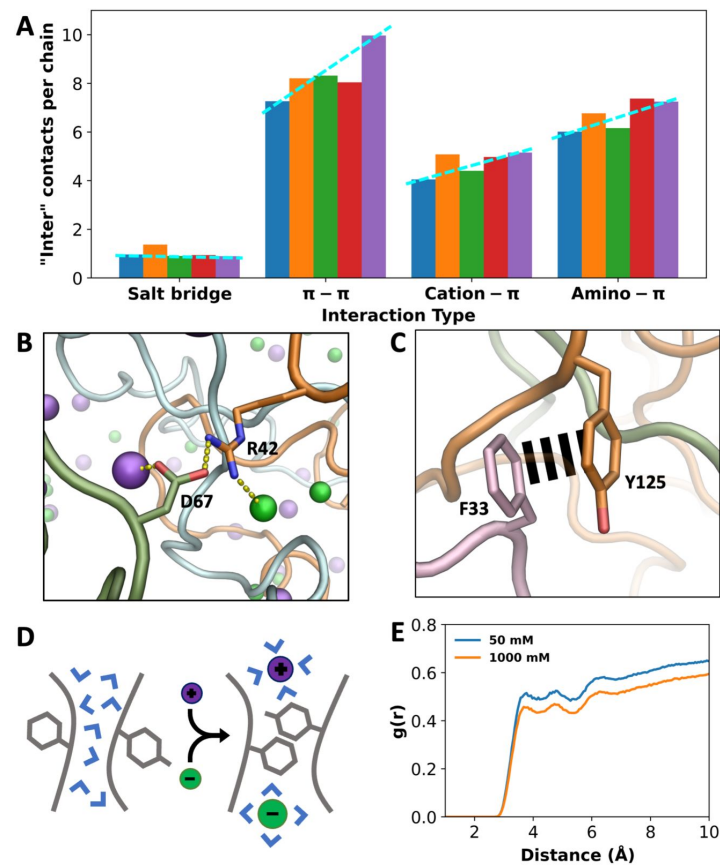


Figure 5.

Indirect effects of ions on different types of interactions. (A) Number of inter-chain contacts per chain for each interaction type. Bars from left to right correspond to increasing salt concentrations (50, 150, 300, 500, and 1000 mM). Dashed lines are drawn to show trends. (B) An inter-chain salt bridge, with ion coordination by the partner sidechains. (C) An inter-chain π - π interaction, free of ion coordination. (D) Schematic showing a π - π interaction facilitated by high salt, via drawing water away from the interaction partners. (E) Radial distribution functions of water around Tyr residues that interact with Phe, Arg, Lys, Gln, and Asn. Lower values at high salt demonstrate water withdrawal.

The weakening of salt bridges by salt can be understood as ions can compete with the salt-bridge partners for coordination (Supplementary Figures 2A and 3A). Indeed, at 1000 mM NaCl, cationic and anionic partners in a salt bridge are often found to also coordinate Cl^- and Na^+ , respectively (Figure 5B). By contrast, π - π interactions are hardly affected by ion competition (Figure 5C), as Tyr has only a weak ability to coordinate Cl^- and Na^+ (Supplementary Figures 2B and 3B) and Phe completely lacks this ability. In comparison, one partner in both cation- π and amino- π interactions can potentially suffer from ion competition (Supplementary Figures 2A and 3A), which may explain why the increases in cation- π and amino- π interactions at high salt are not as large as the counterpart in π - π interactions.

For π -types of interactions, instead of a weakening mechanism through competition, salt may exert a strengthening mechanism by drawing water away from the interaction partners (Figure 5D). We calculated the RDFs of water around Tyr sidechains that form π -types of interactions and found decreases in water density when the salt concentration is increased from 50 to 1000 mM (Figure 5E and Supplementary Figure 4A-C). As a control, no decrease in water density is seen around Asp sidechains that form salt bridges (Supplementary Figure 4D). Water can interfere with and weaken π -types of interactions; by drawing water away from the interaction partners, high salt strengthens these interactions and thereby indirectly contributes to condensation.

Discussion

We have shown in atomistic detail the actions of ions in the condensation of A1-LCD over a wide range of NaCl concentrations. The MD simulations reveal that NaCl has both direct and indirect effects in driving phase separation. The first direct effect is to neutralize the net charge of A1-LCD and thereby attenuate net charge repulsion. The second direct effect is to bridge between protein chains and thereby fortify intermolecular interaction networks. In addition, high salt strengthens π -types of interactions by drawing water away from the interaction partners, thereby also indirectly driving phase separation. The net result is that, while phase separation of A1-LCD is prevented by net charge repulsion at low salt, it is enabled at intermediate salt through charge neutralization and chain bridging by ions. The drive for phase separation becomes even stronger at high salt, where π -types of interactions are strengthened.

These findings broaden our understanding of the roles of charges and ions in phase separation. That high net charge is a suppressive factor is highlighted by the strong effect of A1-LCD charge mutations on C_{sat} .²⁹ Here we have shown that this suppressive action can be countered by the addition of salt, which exerts both direct and indirect effects. One of these direct effects, i.e., neutralization of net charge, can be treated by a Debye-Hückel potential in coarse-grained simulations.³⁰ However, our all-atom explicit-solvent simulations have revealed not only an additional direct effect, i.e., bridging between protein chains, but also an indirect effect, i.e., strengthening of π -type interactions. Krainer et al.¹⁹ observed a reentrant salt effect on the phase separation of IDPs, and attributed the reemergence of phase separation at high salt to strengthened π -type and hydrophobic interactions that overcompensate weakened electrostatic attraction. Their conclusion was based on the salt dependences of the potentials of mean force calculated for pairs of sidechains of various types. Instead of a single pair of sidechains, our simulations are on multiple copies of protein chains. We show that, at high salt, the number of salt bridges is reduced while the numbers of π -types of interactions are elevated in the multi-chain system. The simulations further reveal that high salt achieves the latter effect by drawing water away from π -interaction partners.

The present work puts us in a position to paint a unified picture of salt dependences of homotypic phase separation (Figure 6A, B and Table 1). The salt dependence that is most often reported is screening, where salts weaken the electrostatic attraction between protein chains and thereby suppress phase separation.^{10,17} The reentrant salt dependence reported by Krainer et al.¹⁹

can be seen as an extension of the screening scenario, whereby high salt overcompensates the screening effect by strengthening π -type and hydrophobic interactions and leads to reemergence of phase separation. The high net charge scenario represented by A1-LCD is similar to the reentrant scenario at high salt but differs from it at low salt, where phase separation is prevented by net charge repulsion.^{10,20,25,27} In this class of salt dependence, a certain amount of salt is required to start phase separation. For A1-LCD, this minimum salt is ~100 mM NaCl.¹⁰ The recombinant mussel foot protein-1 (RMFP-1) presents an extreme example, which has a net charge of +24 over a sequence length of 121 and requires up to 700 mM NaCl to begin phase separation.²¹ The fourth class of salt dependence is a variation of the preceding one; here the net charge is low and no significant net charge repulsion is expected, so the protein can phase separate even without salt. For example, FUS-LCD has a low net charge of -2 and phase separates without salt.²⁶ The difference between the high and low net charge scenarios is well captured by the phase-separation behaviors of TDP43-LCD at two pHs.²⁷ At pH 7, this protein has a relatively low net charge of +5 and phase separates without salt; the salt dependence thus belongs to the low net charge class. When pH is lowered to 4, the six His residues in the purification tag become positively charged, raising the net charge to +11; now phase separation requires 300 mM NaCl and the salt dependence switches to the high net charge class.

We can not only rationalize the four distinct classes of salt dependence but also start to predict them from the amino-acid composition of the protein (**Figure 6C** and **Table 1**). The foregoing physical understanding suggests three determinants of salt dependence class: (1) the total number of charged residues, which determines the contribution of electrostatic interactions to the drive for phase separation and the importance of salt screening; (2) the net charge, which determines the magnitude of net charge repulsion; and (3) the total number of aromatic residues, which determines whether phase separation reemerges at high salt. We predict that the screening class occurs when the charged content is high but the net charge and aromatic content are low (**Figure 6C**). The other three classes of salt dependence all require a high aromatic content. For the reentrant class, the increase in aromatic content is the only difference from the screening class. For the high net charge class, the net charge obviously has to be high, but that could also mean at least a moderately high charged content. Finally the low net charge class is predicted when the net charge is low but no bias is required of the charged content. The amino-acid compositional data in **Table 1** validate these predictions. For example, of the 16 proteins in the classes of salt dependence that call for a high aromatic content, 11 (or 68%) actually have this feature (aromatic content > an 8% threshold). In contrast, of the 9 proteins in the class of salt dependence (i.e., “screening”) that calls for a low aromatic content, only 2 (or 22%) have an aromatic content above the threshold. In the latter two cases it remains to be tested whether phase separation would reemerge at much higher salt (and thereby resulting in a reclassification to reentrant). Likewise, of the 9 proteins in the class of salt dependence (i.e., “high net charge”) that calls for a high net charge, 6 (or 67%) actually have this feature (net charge > a 6% threshold). In contrast, of the 16 proteins in the classes of salt dependence that call for a low net charge, none has a net charge above the threshold. Lastly, of the 14 proteins in the classes of salt dependence that call for a high charge content, 10 (or 71%) actually have this feature (charged content > a 20% threshold). In comparison, of the 11 proteins in the classes of salt dependence that do not require a high charge content, only 2 (or 18%) have a charge content above the threshold.

To conclude, salts regulate intermolecular interactions in biomolecular condensates in a variety of ways, but the net effect on phase separation can be placed into four distinct types (**Figure 6A, B**). For homotypic condensates, salt effect types are predictable from the protein amino-acid composition. Salts also exert significant effects on heterotypic condensates;^{8,18} the conclusions drawn here on homotypic condensates may prove instructive for heterotypic condensates. The fact that different IDPs respond to salts differently raises interesting questions on phase separation inside cells. For example, a fluctuation in intracellular salt concentration may promote the phase separation of some IDPs and suppress the phase separation of other IDPs. If two IDPs co-phase separate or form heterotypic condensates, salt fluctuations may change the

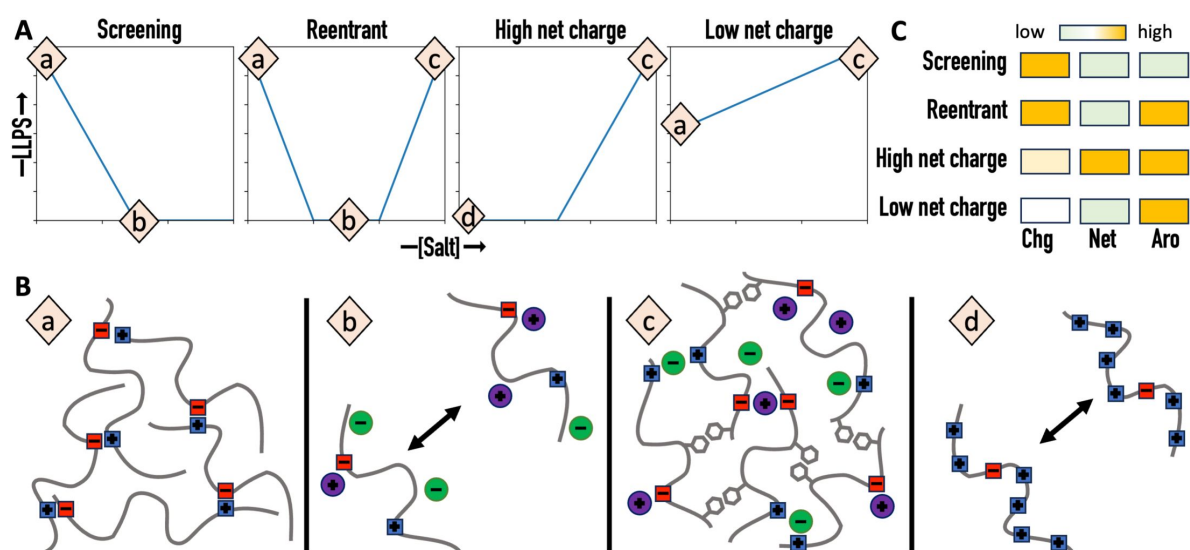


Figure 6.

Four classes of salt dependence and their prediction from amino-acid composition. (A) Salt dependences of liquid-liquid phase separation (LLPS). (B) Charge-charge and π -type interactions and their regulation by salt. (a) Significant Charge-charge attraction. (b) Screening of charge-charge attraction by salt. (c) Strengthening of π -type interactions by high salt. (d) Repulsion due to high net charge. (C) Distinctions of the four classes of salt dependence by three determinants: charged content ("Chg"), net charge ("Net"), and aromatic content ("Aro").

Protein	Length	Charges (+/-/net) ^b	Aromatic ^d	Salt (mM)	Ref
Screening					
hnRNPA1	314	42 / 34 / 8	33	NaCl 50-300	10
FMRP-LCD	188	37 / 28^c / 9	6	NaCl 0-150	11
Oleo30G	139	15 / 12 / 3	7	NaCl 35-280	12
LAF-1	708	86 / 88 / -2	55	NaCl 125-400	13
LAF-1 RGG	191	28 / 22 / 6	14	NaCl 125-300	13
HP1 α	206	33 / 41 / -8	14	NaCl 25-150	14
Ddx4	236	32 / 36 / -4	22	NaCl 100-500	15
PolyQ	729	38 / 34 / 4	35	NaCl 0-150	16
FIB-1	352	55 / 34 / 21	24	NaCl 50-250	17
Reentrant					
FUS	526	51 / 37 / 14	52	KCl 50-2700	19
TDP-43	414	40 / 44 / -4	36	KCl 50-2700	19
Brd4	1362	175 / 150 / 25	48	KCl 50-2150	19
Sox2	317	34 / 21 / 13	19	KCl 50-2150	19
A11	505	51 / 50 / 1	41	NaCl 22.5-500	19
High net change					
A1-LCD	131	12 / 3 / 9	18	NaCl 50-300	10
Lysozyme ^a	129	17 / 9 / 8	12	NaCl 514-1198	20
RMFP-1	121	24 / 0 / 24	24	NaCl 100-500	21
UBQLN2	624	31 / 40 / -9	22	NaCl 100-300	22
UBQLN2 (450-624)	175	4 / 8 / -4	5	NaCl 50-200	22
HBP-2 (pH 5.5)	193	30^c / 13 / 17	20	NaCl 50-500	23
Prp-LCD	122	11 / 1 / 10	11	NaCl 150-750	24
RLP3 ₈	120	12 / 8 / 4	8	NaCl 0-2000	25
TDP43-LCD (pH 4)	148	14 ^c / 3 / 11	12	NaCl 0-300	27
Low net charge					
FUS-LCD	136	0 / 2 / -2	20	NaCl 0-250	26
TDP43-LCD (pH 7)	148	8 / 3 / 5	12	NaCl 0-300	27

^aSingle-domain folded protein.

^bCharges are listed as the number of positively (R and K, including H when specifically indicated) or negatively (D and E) charged residues or net charge.

^cH is treated as positively charged at the low pH condition.

^dAromatic residues are W, Y, and F.

^eHigh contents of charged residues, net charges, and aromatic residues are indicated by bold letters, with thresholds at 20%, 6%, and 8%, respectively, when measured as percentages of the sequence length.

Table 1.

Correlation between class of salt dependence and amino-acid composition

protein composition of condensates and the relative contributions of the individual proteins to phase separation. Lastly, we note that the atomistic details of ion coordination to A1-LCD backbone or sidechain groups are reminiscent of those observed in other systems, in particular channels and transporters for Na^+ and Cl^- .³⁷⁻³⁹

Computational Methods

Molecular dynamics simulations

MD simulations were performed using AMBER 18⁴⁰ with ff14SB force field⁴¹ for the protein and TIP4P-D for water.⁴² The initial configuration of the 8-chain system was constructed using A1-LCD conformations from previous single-copy simulations.³⁶ Specifically, 4 copies with different conformations were placed in a rectangular box, positioned with minimal inter-chain contacts within 3.5 Å. This 4-copy subsystem was duplicated to form the initial configuration of the 8-chain system (**Figure 1B**). The protein chains were solvated in a box with dimensions of 182 Å $\times$ 146 Å $\times$ 164 Å. For each desired salt concentration (50, 150, 300, 500, or 1000 mM), an appropriate number of water molecules were randomly selected and replaced with Cl^- and Na^+ ions; excess Cl^- was added to neutralize the system. The total number of atoms was $\sim$ 500,000 for the 8-chain system at each salt concentration.

After energy minimization in sander (2000 steps of steepest descent and 3000 steps of conjugate gradient), each system was heated to 300 K over 100 ps with a 1 fs timestep, under constant NVT using the Langevin thermostat⁴³ with a 3.0 ps⁻¹ damping constant. The simulation was then continued in four replicates at constant NPT for 1 μ s with a 2 fs timestep. Pressure was regulated using the Berendsen barostat⁴⁴ with a coupling constant of 2.0 ps. All simulations were run on GPUs using *pmemd.cuda*.⁴⁵ Bond lengths involving hydrogens were constrained using the SHAKE algorithm.⁴⁶ Long-range electrostatic interactions were treated by the particle mesh Ewald method.⁴⁷ A cutoff distance of 10 Å was used for the nonbonded interactions. Frames were saved every 200 ps and the last 700 ns was used for analysis.

Data analysis

D_{max} was calculated using the minmax function in VMD.⁴⁸ The differences between the maximum and minimum values of x, y, and z coordinates of all protein atoms were obtained; the largest of the three difference values was taken as D_{max} . Radius of gyration (R_g) was calculated using the radgyr function in CPPTRAJ.⁴⁹ For inter- and intrachain contacts per residue, the number of other residues in contact with a given residue was found using the distance mask function in CPPTRAJ with a cutoff of 6 Å between heavy atoms. Inter-chain contacts per chain were found in a similar way, and were broken into interaction types based on the residue types of the interaction partners. To characterize chain reconfiguration dynamics, the autocorrelation function of the chain end-to-end distance (Ca-Ca distance between the first and last residues) was calculated. After averaging over the 8 chains and then over 4 replicate simulations at each NaCl concentration, the autocorrelation function was fit to an exponential function; the time constant of the fit was taken to be the reconfiguration time.

Radial distribution functions (RDFs) for ions around protein atoms were calculated in CPPTRAJ using the radial command. These RDF plots were used to select cutoffs for 1st- and 2nd-shell ion binding; these cutoffs were 4.0 and 6.4 Å, respectively, for Cl^- and 3.0 and 5.4 Å, respectively, for Na^+ . The number of 1st-shell or 1st- and 2nd-shell ions interacting with N and O atoms in a given sidechain type, the backbone, or the 8 chains together was calculated using CPPTRAJ. Ions were considered bridging if they interacted with two or more chains (based on 2nd-shell cutoffs) at the same time. Those interacting with an exact number of chains (e.g., 3) were identified when the same ion was found in the interaction lists of that many chains.

RDFs for water molecules around Tyr residues were calculated using VMD. Only Tyr residues that interact with specific residue types (within 6 Å of the 6-carbon ring) were selected: the partner residues were Phe (6-carbon ring; for π - π), Arg and Lys (sidechain N atoms; for cation- π), or Gln and Asn (sidechain N atoms; for amino- π). As a control, PRDFs for water were also calculated around Asp sidechain oxygens that interact with Arg and Lys sidechain nitrogens.

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Editors

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Reviewer #1 (Public Review):

Summary:

The authors examined the salt-dependent phase separation of the low-complexity domain of hnRN-PA1 (A1-LCD). Using all-atom molecular dynamics simulations, they identified four distinct classes of salt dependence in the phase separation of intrinsically disordered proteins (IDPs), which can be predicted based on their amino acid composition. However, the simulations and analysis, in their current form, are inadequate and incomplete.

Strengths:

The authors attempt to unravel the mechanistic insights into the interplay between salt and protein phase separation, which is important given the complex behavior of salt effects on this process. Their effort to correlate the influence of salt on the low-complexity domain of hnRNPA1 (A1-LCD) with a range of other proteins known to undergo salt-dependent phase separation is an interesting and valuable topic.

Weaknesses:

(1) The simulations performed are not sufficiently long (Figure 2A) to accurately comment on phase separation behavior. The simulations do not appear to have converged well, indicating that the system has not reached a steady state, rendering the analysis of the trajectories unreliable.

(2) The majority of the data presented shows no significant alteration with changes in salt concentration. However, the authors have based conclusions and made significant comments

regarding salt activities. The absence of error bars in the data representation raises questions about its reliability. Additionally, the manuscript lacks sufficient scientific details of the calculations.

(3) In Figures 2B and 2C, the changes in the radius of gyration and the number of contacts do not display significant variations with changes in salt concentration. The change in the radius of gyration with salt concentration is less than 1 Å, and the number of contacts does not change by at least 1. The authors' conclusions based on these minor changes seem unfounded.

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Reviewer #2 (Public Review):

This is an interesting computational study addressing how salt affects the assembly of biomolecular condensates. The simulation data are valuable as they provide a degree of atomistic details regarding how small salt ions modulate interactions among intrinsically disordered proteins with charged residues, namely via Debye-like screening that weakens the effective electrostatic interactions among the polymers, or through bridging interactions that allow interactions between like charges from different polymer chains to become effectively attractive (as illustrated, e.g., by the radial distribution functions in Supplementary Information). However, this manuscript has several shortcomings: (i) Connotations of the manuscript notwithstanding, many of the authors' concepts about salt effects on biomolecular condensates have been put forth by theoretical models, at least back in 2020 and even earlier. Those earlier works afford extensive information such as considerations of salt concentrations inside and outside the condensate (tie-lines). But the authors do not appear to be aware of this body of prior works and therefore missed the opportunity to build on these previous advances and put the present work with its complementary advantages in structural details in the proper context. (ii) There are significant experimental findings regarding salt effects on condensate formation [which have been modeled more recently] that predate the A1-LCD system (ref.19) addressed by the present manuscript. This information should be included, e.g., in Table 1, for sound scholarship and completeness. (iii) The strengths and limitations of the authors' approach vis-à-vis other theoretical approaches should be discussed with some degree of thoroughness (e.g., how the smallness of the authors' simulation system may affect the nature of the "phase transition" and the information that can be gathered regarding salt concentration inside vs. outside the "condensate" etc.). Accordingly, this manuscript should be revised to address the following. In particular, the discussion in the manuscript should be significantly expanded by including references mentioned below as well as other references pertinent to the issues raised.

(1) The ability to use atomistic models to address the questions at hand is a strength of the present work. However, presumably because of the computational cost of such models, the "phase-separated" "condensates" in this manuscript are extremely small (only 8 chains). An inspection of Fig.1 indicates that while the high-salt configuration (snapshot, bottom right) is more compact and droplet-like than the low-salt configuration (top right), it is not clear that the 50 mM NaCl configuration can reasonably correspond to a dilute or homogeneous phase (without phase separation) or just a condensate with a lower protein concentration because the chains are still highly associated. One may argue that they become two droplets touching each other (the chains are not fully dispersed throughout the simulation box, unlike in typical coarse-grained simulations of biomolecular phase separation). While it may not be unfair to argue from this observation that the condensed phase is less stable at low salt, this raises critical questions about the adequacy of the approach as a stand-alone source of theoretical information. Accordingly, an informative discussion of the limitation of the authors' approach and comparisons with results from complementary approaches such as analytical

theories and coarse-grained molecular dynamics will be instructive-even imperative, especially since such results exist in the literature (please see below).

(2) The aforementioned limitation is reflected by the authors' choice of using D_{max} as a sort of phase-separation order parameter. However, no evidence was shown to indicate that D_{max} exhibits a two-state-like distribution expected of phase separation. It is also not clear whether a D_{max} value corresponding to the linear dimension of the simulation box was ever encountered in the authors' simulated trajectories such that the chains can be reliably considered to be essentially fully dispersed as would be expected for the dilute phase. Moreover, as the authors have noted in the second paragraph of the Results, the variation of D_{max} with simulation time does not show a monotonic rank order with salt concentration. The authors' explanation is equivalent to stipulating that the simulation system has not fully equilibrated, inevitably casting doubt on at least some of the conclusions drawn from the simulation data.

(3) With these limitations, is it realistic to estimate possible differences in salt concentration between the dilute and condensed phases in the present work? These features, including tie-lines, were shown to be amenable to analytical theory and coarse-grained molecular dynamics simulation (please see below).

(4) In the comparison in Fig.2B between experimental and simulated radius of gyration as a function of [NaCl], there is an outlier among the simulated radii of gyration at [NaCl] ~ 250 mM. An explanation should be offered.

(5) The phenomenon of no phase separation at zero and low salt and phase separation at higher salt has been observed for the IDP Caprin1 and several of its mutants [Wong et al., J Am Chem Soc 142, 2471-2489 (2020) [<https://pubs.acs.org/doi/full/10.1021/jacs.9b12208>], see especially Fig.9 of this reference]. This work should be included in the discussion and added to Table 1.

(6) The authors stated in the Introduction that "A unifying understanding of how salt affects the phase separation of IDPs is still lacking". While it is definitely true that much remains to be learned about salt effects on IDP phase separation, the advances that have already been made regarding salt effects on IDP phase separation is more abundant than that conveyed by this narrative. For instance, an analytical theory termed rG-RPA was put forth in 2020 to provide a uniform (unified) treatment of salt, pH, and sequence-charge-pattern effects on polyampholytes and polyelectrolytes (corresponding to the authors' low net charge and high net charge cases). This theory offers a means to predict salt-IDP tie-lines and a comprehensive account of salt effect on polyelectrolytes resulting in a lack of phase separation at extremely low salt and subsequent salt-enhanced phase separation (similar to the case the authors studied here) and in some cases re-entrant phase separation or dissolution [Lin et al., J Chem Phys 152, 045102 (2020) [<https://doi.org/10.1063/1.5139661>]]. This work is highly relevant and it already provided a conceptual framework for the authors' atomistic results and subsequent discussion. As such, it should definitely be a part of the authors' discussion.

(7) Bridging interactions by small ions resulting in effective attractive interactions among polyelectrolytes leading to their phase separation have been demonstrated computationally by Orkoulas et al., Phys Rev Lett 90, 048303 (2003) [<https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.90.048303>]. This result should also be included in the discussion.

(8) More recently, the salt-dependent phase separations of Caprin1, its RtoK variants and phosphorylated variant (see item #5 above) were modeled (and rationalized) quite comprehensively using rG-RPA, field-theoretic simulation, and coarse-grained molecular dynamics [Lin et al., arXiv:2401.04873 [<https://arxiv.org/abs/2401.04873>]], providing additional data supporting a conceptual perspective put forth in Lin et al. J Chem Phys 2020 (e.g., salt-IDP tie-lines, bridging interactions, re-entrance behaviors etc.) as well as in the

authors' current manuscript. It will be very helpful to the readers of eLife to include this preprint in the authors' discussion, perhaps as per the authors' discretion-along the manner in which other preprints are referenced and discussed in the current version of the manuscript.

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Reviewer #3 (Public Review):

Summary:

This study investigates the salt-dependent phase separation of A1-LCD, an intrinsically disordered region of hnRNPA1 implicated in neurodegenerative diseases. The authors employ all-atom molecular dynamics (MD) simulations to elucidate the molecular mechanisms by which salt influences A1-LCD phase separation. Contrary to typical intrinsically disordered protein (IDP) behavior, A1-LCD phase separation is enhanced by NaCl concentrations above 100 mM. The authors identify two direct effects of salt: neutralization of the protein's net charge and bridging between protein chains, both promoting condensation. They also uncover an indirect effect, where high salt concentrations strengthen pi-type interactions by reducing water availability. These findings provide a detailed molecular picture of the complex interplay between electrostatic interactions, ion binding, and hydration in IDP phase separation.

Strengths:

- **Novel Insight:** The study challenges the prevailing view that salt generally suppresses IDP phase separation, highlighting A1-LCD's unique behavior.
- **Rigorous Methodology:** The authors utilize all-atom MD simulations, a powerful computational tool, to investigate the molecular details of salt-protein interactions.
- **Comprehensive Analysis:** The study systematically explores a wide range of salt concentrations, revealing a nuanced picture of salt effects on phase separation.
- **Clear Presentation:** The manuscript is well-written and logically structured, making the findings accessible to a broad audience.

Weaknesses:

- **Limited Scope:** The study focuses solely on the truncated A1-LCD, omitting simulations of the full-length protein. This limitation reduces the study's comparative value, as the authors note that the full-length protein exhibits typical salt-dependent behavior. A comparative analysis would strengthen the manuscript's conclusions and broaden its impact.

Overall, this manuscript represents a significant contribution to the field of IDP phase separation. The authors' findings provide valuable insights into the molecular mechanisms by which salt modulates this process, with potential implications for understanding and treating neurodegenerative diseases. While the study is well-conducted and clearly presented, further research is needed to validate the findings and explore their broader applicability.

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